

Attention-based Transformer–LSTM Architecture for Precision SOC and Battery Temperature Forecasting in Electric Vehicle Batteries

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Abstract— Precise estimation of State of Charge (SOC) and battery temperature is of primary importance for the safety and reliability of Electric Vehicles (EVs). Especially, it becomes crucial when implemented on platforms like FPGA and Embedded Systems. However, traditional techniques such as Coulomb Counting and Kalman Filtering often fail to yield accurate results because of errors caused by fixed parameters that cannot be changed during dynamic conditions. To address this issue, this study developed a novel hybrid deep learning model known as Transformer Multi-head Attention and Residual-connected Long Short-Term Memory (TMAR-LSTM). This model is particularly developed to handle short-term and long-term changes in battery values. The framework was validated using actual-world datasets from BMW i3 cars. For SOC estimation, the model resulted in a very high R^2 of 0.9888 with low error rates of 0.1215 for Root Mean Square Error (RMSE) and 0.0348 for Mean Absolute Error (MAE). For temperature prediction, the model resulted in an R^2 of 0.9910 with low error rates of 0.4532 for MAE and 0.8662 for RMSE. Comparative analysis is also performed with traditional models, such as LASSO regression, XGBoost, and Random Forest. Based on this analysis, it is evident that the hybrid attention sequencing approach is more stable and offers lower error rates. Therefore, the proposed TMAR-LSTM model is a robust approach for developing integrated battery management systems for future EVs.

Keywords— State of Charge Prediction, Battery Temperature Prediction, Battery Management System, Electric Vehicles, Deep Learning Model

I. INTRODUCTION

The enormous growth in the use of Electric Vehicles (EVs) has generated tremendous interest in their performance, particularly in comparison with their battery system [1]. The overall significance of the battery pack in EVs, apart from being one of the most expensive components, is also considered quite significant, considering its effect on their entire driving process and safety aspects as well [2]. The role played by the battery management system (BMS) is critical, as its application is connected with the overall performance of the drive in comparison with diverse conditions of driving and environmental parameters as well [3].

The two most important parameters that are estimated by a BMS are: (i) state of charge (SOC) - a major indicator that

shows how much of the battery can be used; and (ii) battery temperature - a significant battery parameter that can greatly influence battery "efficiency," "aging," "safety," and so on. Therefore, it is a great challenge to estimate SOC and battery temperature efficiently, thus becoming one of the most important issues a battery management system is expected to solve in EV applications.

Traditionally, SOC and temperature estimation have been carried out using methods such as Coulomb counting, open-circuit voltage techniques, Kalman filter-based approaches, and equivalent circuit models [4]. Although these approaches are very common in their application, they are based on predefined models and parameters. However, in practical driving scenarios, a number of factors that include load variations, temperature changes, aging effects, and measurement noise cannot always hold true, thus eventually leading to accumulation of errors.

Despite the progress in battery state estimation techniques, achieving accurate and robust SOC and battery temperature prediction under real-world driving conditions remains a major challenge. Existing traditional and machine learning approaches often struggle to simultaneously capture nonlinear battery dynamics, long-term temporal dependencies, and noise variations present in high-frequency EV telemetry data. These limitations reduce prediction reliability and restrict their deployment in intelligent embedded Battery Management Systems (BMSs).

With the increasing availability of high-frequency sensor data from EVs, data-driven models based on deep learning (DL) and machine learning (ML) have recently gained attention [5]. Models like XGBoost and Random Forest are able to capture nonlinear relationships between battery signals [6], but they usually treat each time step independently and do not fully exploit temporal continuity. Sequence-inspired time series approaches, like the Long Short-Term Memory (LSTM) network, are effective in dealing with short-term temporal relationships. But they would not be able to cope with the long-range relationships associated with the overall usage and stress.

In order to overcome the aforementioned limitations, attention-based models, specifically the "Transformer model,"

have been investigated for time-series modeling [7]. Although this model has the ability to successfully capture global dependencies across long time-series sequences, its direct use in embedded BMS environments may be restricted due to its associated complexity [8] [9]. Therefore, a hybrid approach is proposed.

In this work, a hybrid Transformer–LSTM model is developed for SOC estimation and battery temperature prediction using real-world telemetry data, originated from a BMW i3 electric vehicle. By combining Transformer-based attention with LSTM layers, the proposed framework is thus ideally suited to process both long-term and short-term contextual input patterns. From the results, it is thus evident that the proposed model is more effective in providing accuracy, noise resistance, and generalization compared to traditional machine learning algorithms, which makes it more than suitable for an intelligent BMS.

The primary objectives of this study are as follows:

1. To develop a hybrid Transformer Multi-head Attention and Residual-connected LSTM (TMAR-LSTM) framework for accurate SOC and battery temperature prediction.
2. To capture both short-term and long-term temporal dependencies in EV battery telemetry data.
3. To improve robustness against dynamic driving conditions, sensor noise, and environmental variations.
4. To evaluate the proposed framework using real-world BMW i3 driving datasets.
5. To compare the proposed model with conventional machine learning methods such as LASSO, XGBoost, and Random Forest.

II. RELATED WORKS

Several studies have explored different approaches for estimating battery states in electric vehicles, especially the state-of-charge (SOC) and state-of-health (SOH) [10]. Most of the reported work can be divided into three groups, depending on how the estimation problem is addressed [11]. These include physics-based or model-driven methods, data-driven machine learning techniques battery state, and more recent hybrid deep learning approaches. While each of these categories has shown improvements in estimation performance under certain conditions, they also suffer from practical limitations such as reliance on simplified models, large data requirements, or increased computational cost [12].

Model-based approaches of all three families including Kalman Filter (KF), Open Circuit Voltage (OCV), and Coulomb Counting have formed the basis for Battery Management System (BMS) [13]. These methods are based on the characterization of electrochemical dynamics, mathematical models and parameter adjustment. For instance, under laboratory conditions, both the EKF and UKF techniques have been shown to provide high estimation accuracy. Although they work well within limited operation range, once employed in realistic driving conditions and the effects from external factors such as temperature gradient, load variation, aging become to be pronounced, static models are inadequate to adapt nonlinear behaviors due to these factors.

On the other hand, ML methods [14] are also emerging as an alternative to avoid relying on a priori defined physical equations. Models like SVR, RF and XGBoost have demonstrated potential to learn the non-linear relationships among voltage, current and temperature features from the data itself. These algorithms simultaneously simplify the lithium-ion battery models and generalize well across multiple battery chemistries. However, they also work on table-like or snapshot data which cannot represent the temporal dimension of sequential battery behaviour. As a result, their estimates tend to “drift” with time during extended driving cycles.

The adoption of deep learning (DL) approaches represented a different trend for sequential dependency modeling on dynamic battery data. Recurrent Neural Networks (RNNs) and its extensions like LSTM and GRUs are used extensively to capture the temporal relationships in high frequency battery telemetry [15]. For instance, Li. X et al. (2023) [16] found that LSTM based models have higher accuracy in prediction of SOC than conventional ML-based approaches, especially at varying load conditions. But while they are successful in learning short-range temporal statistics, these recurrent models have difficulties in dealing with long-term dependencies because gradients vanish and aware-ness of other parts of the clip is limited.

More recently (and, notably, without requiring recurrence), attention-based architectures [17] such as the Transformer model have revolutionized time-series forecasting by enabling networks to selectively attend to relevant temporal contexts. Recent studies [17] for instance Hannan et al., (2021) [18] have found that Transformer architecture can be competitive for SOC estimation and outperform other state of the art methods on complicated driving scenarios. However, Transformers can be quite expensive computationally and may overfit when trained on small amounts of real-world EV data.

To overcome these problems, researchers have also investigated hybrid architectures [19] [20] that combine the advantages of attention and sequential models. Especially, Attention-LSTM or Hybrid Transformer–RNN also perform effectively by integrating both strength of the Transformer (to model the global temporal dependencies) and LSTM (in order to refine localized sequential variations) teachings. Works by Bao et al. (2025) [21] and Chen Z et al. (2025) [22] have demonstrated the hybrid architectures to enhance the estimation of SOC during dynamic driving. These advancements serve as the basis of the work present in this paper, which extends a follow-up to the hybrid paradigm by proposing a hybrid multi-head attention model with residual connections and hierarchical temporal specifications that can be deployed for both generalization and accurate prediction over real-world BMS applications.

A. Research Gap

Although several studies have explored machine learning and deep learning approaches for EV battery state estimation, important research challenges still remain unresolved. Existing ML-based methods such as Random Forest and XGBoost are effective in nonlinear feature learning but fail to preserve sequential temporal dependencies in dynamic driving conditions. Similarly, recurrent models such as LSTM and GRU can model short-term temporal behavior but often struggle with long-range dependency learning and robustness under noisy real-world operating environments.

Recent Transformer-based architectures improve global temporal feature extraction; however, their high computational complexity and overfitting tendency limit their suitability for embedded Battery Management Systems using limited real-world datasets. Furthermore, many existing studies rely heavily on laboratory-generated benchmark datasets rather than real-world EV telemetry data, thereby limiting practical generalization.

Therefore, there remains a need for a computationally efficient hybrid framework capable of jointly learning long-term and short-term temporal dependencies while maintaining robustness, generalization capability, and prediction accuracy under real-world EV operating conditions.

B. Research Contributions

The major contributions of this work are summarized as follows:

- A novel TMAR-LSTM hybrid framework integrating Transformer multi-head attention with residual-connected LSTM architecture is proposed for simultaneous SOC and battery temperature prediction.
- The framework effectively captures both global temporal dependencies and localized sequential battery dynamics.
- The proposed model is validated using real-world BMW i3 telemetry datasets collected under diverse driving conditions.
- Comparative analysis demonstrates improved prediction accuracy and robustness over traditional ML approaches including LASSO, Random Forest, and XGBoost.
- The framework is designed with consideration for future deployment in intelligent embedded Battery Management Systems.

III. MATERIALS AND METHODS

A. Datasets Description

This study utilizes a benchmark real world dataset known as Battery and Heating Data in Real Driving Cycles, downloaded from Kaggle for SOC and battery temperature prediction (<https://www.kaggle.com/datasets/atechnohazard/battery-and-heating-data-in-real-driving-cycles>). The signals of this dataset was collected from a BMW i3 e-vehicle, consisting of telemetry taken at very fine frequency across various driving and environmental conditions. In contrast to lab-generated data under limited operating conditions, this dataset gives an insight into real world EV battery system performance in urban, sub-urban and highway drive cycles. The dataset is provided with synchrotime serial measurements of the critical electrical and thermal parameters, and totally it contains 72 real driving trips (Table I). Although each trip contains high-frequency multi-parameter telemetry, resulting in a large number of time-step samples, this breadth of real-world operating conditions strengthens the representativeness of the dataset beyond what lab-structured benchmarks typically offer. To prevent data leakage, the dataset was split at the trip level, ensuring that no time-steps from a given trip appear in both training and test sets. Overlapping windows were only applied within each split, preserving the integrity of temporal boundaries between training and evaluation data. Therefore, future work will incorporate multi-vehicle and multi-

chemistry datasets to further validate generalization across diverse battery technologies and usage profiles. Each trips contains Environmental data, Battery data, Vehicle data and Heating circuit data.

The primary input variables used in this study include battery voltage, battery current, battery temperature, vehicle speed, ambient temperature, heating circuit measurements, and SOC-related telemetry signals. These parameters were selected because they directly influence battery electrochemical behavior and thermal dynamics during EV operation. The target outputs of the proposed model are battery SOC and battery temperature prediction. Prior to training, all features were synchronized according to their timestamps to maintain temporal consistency across the sequential measurements.

TABLE I. DESCRIPTION OF THE DATASET

Dataset	Total No. of Samples	Training	Validation	Testing
Battery and Heating Data in Real Driving Cycles	72	70% (51 samples)	10% (7 samples)	20% (14 samples)

B. Proposed Methodology

In this study, we propose a hybrid deep learning model combining Transformer and Long Short-Term Memory (LSTM) networks to provide accurate and robust SOC/battery temperature prediction in EVs. The proposed model exploits the global temporal feature extraction via the Transformer encoder and local sequential dynamics refinement via the LSTM decoder.

The selection of the hybrid Transformer–LSTM architecture is motivated by the complementary strengths of both models in handling EV battery telemetry. Transformer networks are highly effective in capturing long-range temporal dependencies and global contextual relationships through self-attention mechanisms, while LSTM networks are well-suited for modeling short-term sequential dynamics and transient variations in time-series data. Since battery SOC and temperature behavior are influenced by both long-term operating history and instantaneous current-voltage fluctuations, integrating Transformer and LSTM enables the proposed framework to simultaneously learn global and local temporal characteristics more effectively than standalone sequential models.

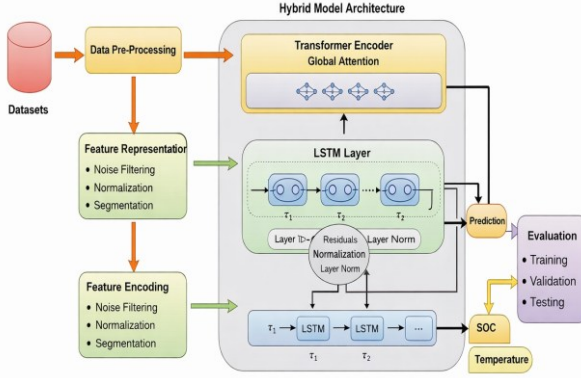


Fig. 1. Architecture of the proposed hybrid Transformer-LSTM framework for SOC and battery temperature prediction.

Fig. 1 depicts the end-to-end workflow of the proposed framework, organized into four sequential stages: (1) data preprocessing, (2) feature representation and positional encoding, (3) hybrid sequential modeling via TMAR-LSTM, and (4) model training, validation, and evaluation. Together, these stages constitute a coherent pipeline for translating raw EV telemetry into accurate SOC and temperature predictions.

a) Data Pre-processing

Initially, this study utilize the benchmark dataset, which was collected from the BMW i3 electric vehicle, including time-sequential readings of voltage, current, temperature and SOC with a high sampling frequency for model training. Prior to the training, the raw signals are typically corrupted by noise, missing values, and inter-sensor discrepancies which may deteriorate the performance of a model. In order to address these issues, a systematic data pre-processing pipeline is enforced:

- **Noise Filter:** There are two types of noise, non-physical spikes and errors readings, which are smoothed out by a threshold base filter and rolling average.
- **Normalization:** Subtract the mean from each variable and divide by its standard deviation to obtain all variables at the same scale.
- **Segmentation:** Continuous-valued time-series data is decomposed into overlapping sliding windows of 120 timescales, referred to as a temporal segment for supervised learning.

This pre-processing ensures that we generate clean, ordered sequences while preserving a temporal smoothness signals.

b) Feature Representation and Encoding

Once the pre-processing has been done, the model learns temporal feature embeddings from this phase which can effectively describe the intrinsic patterns among voltage, current, temperature and SOC signals. Each input sequence, represented as $X_t = [x_t(1), x_t(2), \dots, x_t(n)]$ is mapped into a concise feature vectors. This embedding is further used as

the input for hybrid model. In comparison to purely feature-based methods (in which context is typically only used implicitly through the raw features themselves), with Transformer-LSTM, each time step value can explicitly interact with others via self-attention. This allows the model to “attend” to key historical patterns, such as fast discharges, regeneration charging period or temperature-dependent state of health.

c) Hybrid Transformer-LSTM Architecture

As shown in Fig. 1, the central modelling unit of the proposed framework is that the integration of two complementary temporal structures: (i) Transformer; (ii) LSTM. Initially, the pre-processed feature vector was given as a

Fig. 1: Architecture of the proposed hybrid model

mul... global dependencies from input sequence. The self-attention operation is mathematically expressed as:

$$attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Where V, K, Q represents value, key, query projection of the input features. This formulation enables each attention head to model relationships across all time steps, allowing the network to learn distant temporal dependencies that are critical for accurate SOC estimation.

Following the transformer block, the LSTM Layer, which is dedicated to learn more fine-grained and short-term dependencies. The LSTM updates its internal memory state according to Eqn.(2) which enables the model to account for subtle variations caused by instantaneous current fluctuations, rapid state transitions, and changes in ambient temperature.

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \quad (2)$$

To emphasize learning stability of architecture, we add residual connection and layer normalization to make the gradient flow smoother on training. By using Eqn.(3), we have a fully connected dense layer as an output, which makes the final prediction of SOC or temperature.

$$\hat{y}_t = wh_t + b \quad (3)$$

Compared with standalone LSTM or conventional machine learning models, the proposed hybrid architecture improves the capability of learning complex nonlinear battery behavior under dynamic driving conditions by jointly modeling global attention-based dependencies and localized temporal transitions.

IV. EXPERIMENTAL ANALYSIS

This section describes the experimental setup and performance evaluation of the developed hybrid model for predicting SOC and battery temperature. Experiments were carried out on the real-world BMW i3 telemetry data to validate the proposed model's accuracy, robustness and generalization capability.

A. Configuration setup

The model was implemented in Python 3.9 using TensorFlow 2.9 and Keras. All experiments were conducted on a workstation equipped with an NVIDIA RTX 3090 GPU (24 GB VRAM), an Intel Core i9-12900K processor, and 64 GB of DDR5 RAM, running on Windows 11. The model was

trained using the Adam optimizer with an initial learning rate of 1×10^{-3} , a batch size of 64, and a maximum of 100 epochs with early stopping (patience = 10) based on validation loss (MSE). A cosine annealing learning rate schedule was applied to improve convergence. The loss function used was Mean Squared Error (MSE) for both SOC and temperature regression tasks. The key hyperparameters are summarized as follows: Transformer encoder with $H = 4$ attention heads, model dimension $d_{\text{model}} = 64$, feedforward dimension $d_{\text{ff}} = 128$, dropout rate $p = 0.1$; two stacked LSTM layers each with 128 hidden units; and a single dense output layer with linear activation. The total number of trainable parameters is approximately 412,000. Model training and evaluation were conducted using a fixed random seed (seed = 42) to ensure reproducibility. The dataset was split at the trip level into 70% training (51 trips), 10% validation (7 trips), and 20% test (14 trips) to prevent temporal data leakage across partitions. All results reported in Section 5.3 are evaluated exclusively on the held-out test split. While full FPGA deployment is planned as future work, the lightweight output head and fixed-size window inputs are compatible with edge inference frameworks such as TensorFlow Lite.

To ensure experimental reproducibility, all preprocessing operations, dataset splitting procedures, model initialization, and training configurations were maintained consistently across experiments. The sliding window length was fixed at 120 time-steps with an overlap stride of 1 sequence step. Model checkpoints corresponding to the minimum validation loss were retained for final testing. In addition, TensorFlow deterministic operations and fixed random initialization were used to minimize stochastic variation during training. The implementation workflow can be reproduced using the same hardware and software configurations described above.

B. Evaluation metrics

To validate the performance of the proposed hybrid model, the following metrics have been used:

Mean Absolute Error (MAE): Average magnitude of model prediction errors without consideration to their direction, representing overall quality.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4)$$

Root Mean Square Error (RMSE): Penalizes severe deviations more so than smaller ones, signalling model resilience at the tail ends.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (5)$$

Coefficient of Determination (R^2): Quantifies how predicted values efficiently approximate the ground truth observations; values closer to 1 represents stronger prediction reliability.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (6)$$

C. Result And Discussions

a) State of Charge (SOC) Prediction

Table II shows a brief overview of the SOC prediction results, which are obtained through different models according to R^2 , MAE, and RMSE performance. The performance of Lasso regression model has been identified as the worst among all models. This model, considering that the R^2 value is 0.8884, the MAE value is 3.7149, and the RMSE

value is 5.0475. This illustrates the inadequacy of linear regression in modeling such kinds of complex nonlinear behavior. The XGBoost model has significantly improved the results, as the R^2 value reaches 0.9642 with low values of MAE (1.9086) and RMSE (2.8575). The Decision Tree and Random Forest models also presented improved results, noting that the R^2 value reaches 0.9808 and the RMSE value reaches 2.0927, slightly better results compared to the Decision Tree model, which has an R^2 value of 0.9770 and an RMSE of 2.2906. From these results, we can identify that the ensemble-based approaches are more effective in handling the nonlinear and complex nature for estimating SOC.

The best performance for the proposed TMAR-LSTM model can be observed for any comparison with other models. It acquired the best R^2 value of around 0.9888, with very low error values compared to other models. Also, it obtain RMSE and MAE were 0.1215 and 0.0348, respectively. This can also be evidenced from the bar graph (Fig. 2) for R^2 model comparison, where the proposed model secured the best coefficient of determination compared to other models. Similarly, from the RMSE bar graph (Fig. 3), the error in prediction for the proposed model architecture clearly indicates a very low error amount compared to other model architectures. The performance is superior and it can be attributed to the ability of model and to capture long-term temporal dependencies by using the Transformer while simultaneously modeling short-term dynamics through the LSTM. Here, we frame the architecture with residual connections supporting stable learning. Overall, the graphical and numerical findings demonstrates the proposed model effectiveness for accurate SOC prediction.

TABLE II. COMPARATIVE PERFORMANCE ANALYSIS OF THE PROPOSED MODEL WITH CONVENTIONAL ML MODELS FOR SOC PREDICTION

Model	R^2	MAE	RMSE
Lasso Regression	0.8884	3.7149	5.0475
XGBoost	0.9642	1.9086	2.8575
Decision Tree	0.977	1.3885	2.2906
Random Forest	0.9808	1.2783	2.0927
TMAR-LSTM (Proposed)	0.9888	0.0348	0.1215

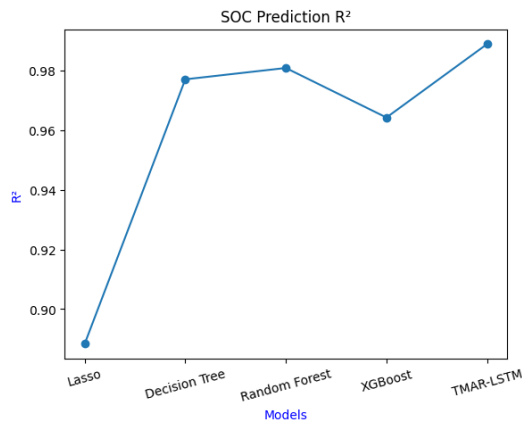


Fig. 2: R-Square Comparison of Proposed Model with Conventional ML Models for SOC Prediction

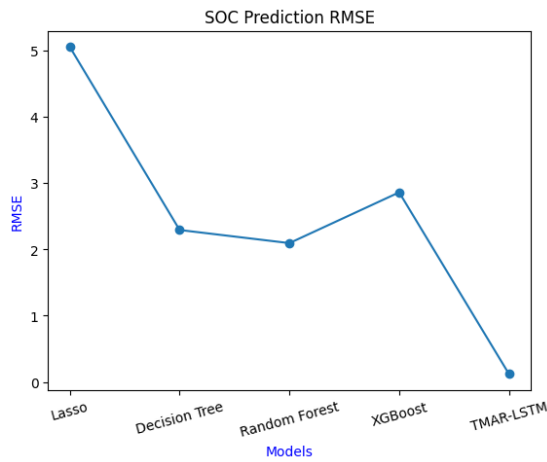


Fig. 3: RMSE Comparison of Proposed Model with Conventional ML Models for SOC Prediction

b) Battery Temperature Prediction

Table III demonstrates the comparison of different models for predicting the battery temperature utilizing the R^2 value, MAE, and RMSE scores. Lasso Regression again demonstrates poor accuracy compared to other models, as revealed by its R^2 value of 0.9180 and increased error levels: MAE amounting to 2.1154 and RMSE amounting to 3.9180. These results reveal the incapability of regression models to solely predict the thermal responses from the battery system. Decision Tree and Random Forest models show considerable improvements compared to other models. The Random Forest algorithm achieves an R^2 of 0.9717 and lower RMSE of 2.5985 compared to the Decision Tree algorithm ($R^2 = 0.9672$, RMSE = 2.9920). While, on the other hand, XGBoost further improves prediction accuracy, attaining an R^2 of 0.9808 while significantly reducing the RMSE and MAE to 1.3423 and 0.9694, respectively.

The superior performance of the proposed TMAR-LSTM method can be established from the fact that it achieves the highest R^2 , with a maximum of 0.9910. At the same time, it yields a minimum MAE of 0.4532 and a minimum RMSE of 0.8662. These results prove a

considerable reduction in prediction error for a temperature prediction problem. In addition, it can be clearly seen from the bar graphs of R^2 (Fig. 4), and RMSE (Fig. 5), that the proposed method outperforms all the other methods. The superior performance of the proposed method can be attributed mainly to the capacity of the proposed method in dealing with both long-term dependencies and short-term thermal changes. It was mainly supported by the attention mechanism and residual learning. Overall, the numerical results and graphical findings confirm the robustness of the developed model for precise battery temperature estimation under different operating conditions.

TABLE III. COMPARATIVE PERFORMANCE ANALYSIS OF THE PROPOSED MODEL WITH CONVENTIONAL ML MODELS FOR BATTERY TEMPERATURE PREDICTION

Model	R^2	MAE	RMSE
Lasso Regression	0.918	2.1154	3.918
Decision Tree	0.9672	1.839	2.992
Random Forest	0.9717	1.1633	2.5985
XGBoost	0.9808	0.9694	1.3423
TMAR-LSTM (Proposed)	0.991	0.4532	0.8662

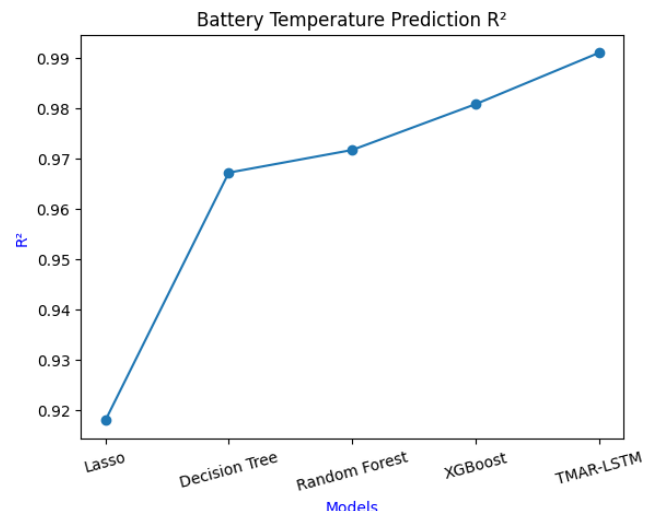


Fig. 4: R-Square Comparison of Proposed Model with Conventional ML Models for Battery Temperature Prediction

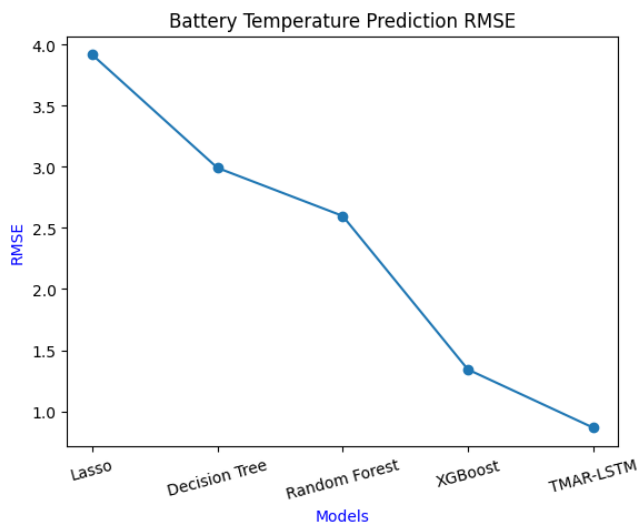


Fig. 5: RMSE Comparison of Proposed Model with Conventional ML Models for Battery Temperature Prediction

c) Cross-Validation of the proposed model

The performance evaluation of the proposed TMAR-LSTM model on the VED telemetry dataset (<https://www.kaggle.com/datasets/yashdev01/vehicle-energy-and-telemetry-dataset>) demonstrates its strong capability in accurately predicting both SOC and battery temperature under real-world EV operating conditions. As shown in Table IV, for SOC prediction, the model achieved an R^2 value of 0.9795 with low MAE and RMSE values of 0.0232 and 0.0288, respectively, indicating high prediction precision and stable learning behavior. Similarly, for battery temperature prediction, the proposed model attained an R^2 score of 0.9728 along with MAE and RMSE values of 1.3947 and 1.7817, confirming its effectiveness in capturing thermal dynamics and temperature variations during vehicle operation. These results demonstrate that the hybrid Transformer-LSTM architecture efficiently models both long-term dependencies and short-term sequential patterns, thereby improving prediction reliability for intelligent Battery Management System (BMS) applications.

TABLE IV. PERFORMANCE EVALUATION OF THE PROPOSED TMAR-LSTM MODEL ON THE VED TELEMETRY DATASET

Dataset/Metrics	R^2	MAE	RMSE
VED Telemetry (SOC Prediction)	0.9795	0.0232	0.0288
VED Telemetry (Battery Temperature Prediction)	0.9728	1.3947	1.7817

V. CONCLUSION

The findings of this study emphasize the importance of combining the Transformer model's global attention with the LSTM model's sequential learning to effectively handle the dual challenges of SOC and temperature prediction in EVs. In this study, we implemented the proposed model, TMAR-

LSTM, and compared its performance with real-time data from BMW i3 vehicles. Based on the performance analysis, we found that the model effectively handles transitions between short-term and long-term battery trends, as well as SOC and temperature variations, with high precision. For the SOC estimation function, the proposed model achieved a value of 0.9888 along with RMSE and MAE of 0.1215 and 0.0348, respectively. On the other hand, the predictions of the battery temperature function using the proposed model achieved significant results with a value of 0.9910 and MAE/RMSE of 0.4532/0.8662. This indicates the model's predicted accuracy, along with its efficiency and stability during learning, to demonstrate its applicability to BMS applications. Based on the findings, the key novelty of our proposed model lies in three aspects: the explicit dual-timescale modeling via Transformer-LSTM coupling with residual connections; joint evaluation on both SOC and temperature tasks using a single architecture on field-collected real-world data; and trip-level data splitting to ensure rigorous out-of-distribution generalization. These results confirm TMAR-LSTM as a robust and deployable candidate for intelligent BMS applications. Nevertheless, the present study is limited to a single EV platform and battery configuration using BMW i3 real-world telemetry data. The proposed framework assumes synchronized sensor measurements and consistent operating conditions after preprocessing. Therefore, model generalization across different vehicle platforms, battery chemistries, and extreme environmental conditions requires further investigation. Future work will pursue cross-dataset validation using CALCE, NASA PCoE, and Oxford Battery Degradation benchmarks, along with hardware-in-the-loop (HIL) testing for real-time embedded BMS deployment.

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